

# Capstone failure testing

Project: Grounded Learning Service

## 1. Prepare the purpose, prerequisites and materials

### Teach the learning cycle

Primary target: Explain the mechanism and terminology of capstone failure testing

#### Session · Sequence and minutes · Resources

**Session:** Session 1 · Understand · 45 min

**Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

**Resources:** Sections 1–6; record the brief recall in the workbook.

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**Session:** Session 2 · Practise · 45 min

**Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

**Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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**Session:** Session 3 · Investigate and transfer · 45 min

**Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

**Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

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This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

## Retrieval answer guidance

### Capstone service implementation

Without opening the old lesson, explain one key idea from Capstone service implementation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of capstone service implementation

Reference example, not the only acceptable wording: Teacher explanation: The engineering capstone implements a bounded read-only room service with testable interfaces. Start with a deterministic baseline before optional model wording. Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate output and log minimised metadata. Keep side-effect tools absent.

### Red team exercises

Without opening the old lesson, explain one key idea from Red team exercises. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of red team exercises

Reference example, not the only acceptable wording: Teacher explanation: Red-team exercises probe explicit failure boundaries in an authorised isolated fixture environment. Their purpose is evidence-based improvement. Use fictional records and allowed test routes. Define a target contract violation, record the exact case, observe action/data traces and preserve a reproducible report. Do not touch external systems.

### Accessibility and user feedback

Without opening the old lesson, explain one key idea from Accessibility and user feedback. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of accessibility and user feedback

Reference example, not the only acceptable wording: Teacher explanation: Accessibility makes information and controls usable across stated needs. Feedback should describe the problem and recovery action without exposing sensitive details. A form provides visible labels, keyboard access, associated error text and a non-colour status indicator. Test at zoom and with keyboard. Do not treat automated checks as a complete user evaluation.

Two 45-minute sessions. Capstone failure testing checks denied roles, missing evidence, invalid input, timeouts and stale data. Each expected outcome is written before the run. Use the supplied constructed scenarios to make the mechanism inspectable. Collect W4 before sharing solutions. Students should name intermediate evidence and limitations; classroom pacing and learning effectiveness remain unvalidated.

### Outcomes

Explain the mechanism and terminology of capstone failure testing

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

## Preparation

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

## Prerequisite answer

A result depends on the inputs, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from predictions or unmeasured conditions. If the prior concept cannot be explained, use a short retrieval activity before this chapter.

## End goal for the pictorial case

Explain capstone failure testing using the deciding step in this fictional service.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

# 2. Explain and model the mechanism

Capstone failure testing checks denied roles, missing evidence, invalid input, timeouts and stale data. Each expected outcome is written before the run.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Use a failure matrix linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

### Term · Meaning

**Term:** Failure matrix

**Meaning:** Conditions with expected results and forbidden operations

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**Term:** Trace assertion

**Meaning:** A check on actual operations

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**Term:** Regression fixture

**Meaning:** A preserved case rechecked after change

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## Capstone failure testing — evidence trace

### Input and scope · Check or calculation · Conclusion and limit

**Input and scope:** Four cases: malformed request, student teacher-guide request, missing register, valid current record. Two deny, one reviews, one supports; no restricted reads occur.

**Check or calculation:** Use a failure matrix linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.

**Conclusion and limit:** All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures. Qualification: A case passes only all its declared response and trace requirements, not merely a convenient status code. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

## Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Four cases: malformed request, student teacher-guide request, missing register, valid current record. Two deny, one reviews, one supports; no restricted reads occur.

All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

## Thinking move · Teacher prompt

**Thinking move:** Plan

**Teacher prompt:** What is given? What is the question? Which procedure fits these facts?

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**Thinking move:** Do

**Teacher prompt:** Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

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**Thinking move:** Check

**Teacher prompt:** Does the output answer the question? Which assumption or limitation prevents a larger claim?

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Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

## A pictorial worked case: Capstone failure testing

Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval.

## The facts we will use

F4 records club time 15:00. The contract permits a checked fixed-template answer when drafting is unavailable.

## Capstone failure testing

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Verify source

Retrieval supplies F4 and the time 15:00.



#### 2. Inject failure

Disable drafting without changing source evidence.



#### 3. Use bounded fallback

Return the checked template citing F4 and record degraded operation.

### CHANGE ONE THING

Make retrieval unavailable as well.

Explain the changed result using the matching lesson.

### Read the picture one step at a time

#### Step · What happens and why

**Step:** 1. Verify source

**What happens and why:** Retrieval supplies F4 and the time 15:00.

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**Step:** 2. Inject failure

**What happens and why:** Disable drafting without changing source evidence.

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### Step: 3. Use bounded fallback

**What happens and why:** Return the checked template citing F4 and record degraded operation.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

Failure injection changes a component deliberately so the team can inspect the response. A fallback must retain evidence checks and honest status.

### What this example does not establish

This trial covers one failure combination; check other components and interactions separately.

### Change one thing in the pictured case

Change: Make retrieval unavailable as well.

Prediction: Return unsupported or unavailable; do not use the factual template.

Reason: A template can phrase evidence but cannot replace missing evidence.

### Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

## 3. Demonstrate and guide practice

### Time or stage · Teacher action

**Time or stage:** Session 1: 0–7 minutes

**Teacher action:** Retrieve the prerequisite; identify the input, procedure and evidence source.

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**Time or stage:** 7–18 minutes

**Teacher action:** Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.

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**Time or stage:** 18–32 minutes

**Teacher action:** Trace the declared procedure and worked case. Complete W2, recording intermediate steps.

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**Time or stage:** 32–45 minutes

**Teacher action:** Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

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**Time or stage:** Session 2: 0–8 minutes

**Teacher action:** Retrieve the worked reasoning and collect the guided response before discussing it.

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**Time or stage:** 8–20 minutes

**Teacher action:** Compare a proposed boundary variant with the original case; identify what remains fixed.

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**Time or stage:** 20–35 minutes

**Teacher action:** Collect W4 independently. Require the procedure trace and the bounded conclusion.

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**Time or stage:** 35–45 minutes

**Teacher action:** Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

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Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

## Reduce help deliberately

Completion task: Four cases: malformed request, student teacher-guide request, missing register, valid current record. Two deny, one reviews, one supports; no restricted reads occur. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

## Individual reasoning check

Use the worked case for Capstone failure testing. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A case passes only all its declared response and trace requirements, not merely a convenient status code.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

## Pictorial practice question

Explain why retrieval success permits the checked 15:00 fallback when drafting fails. What changes if retrieval fails too, and why can a template not fill that evidence gap?

## Reasoned teacher answer

Retrieval supplies F4 and the time 15:00. Disable drafting without changing source evidence. Return the checked template citing F4 and record degraded operation. Changed case: Return unsupported or unavailable; do not use the factual template. A template can phrase evidence but cannot replace missing evidence.

Changed-case reasoning: Return unsupported or unavailable; do not use the factual template. A template can phrase evidence but cannot replace missing evidence.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

# 4. Run the investigation and diagnose misconceptions

The worked result establishes every future case: A case passes only all its declared response and trace requirements, not merely a convenient status code.

A missing input may be replaced without recording the change: A replacement changes the evidence or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent evidence: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

## Match the support to the first error

## What the response shows · Next teaching action

**What the response shows:** Missing or misread facts

**Next teaching action:** Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

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**What the response shows:** Wrong procedure

**Next teaching action:** Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

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**What the response shows:** Arithmetic or implementation error

**Next teaching action:** Trace intermediate values or lines; preserve the first mismatch and retry that step.

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**What the response shows:** Correct result without a reason

**Next teaching action:** Ask for a trace and a changed case before marking independent understanding.

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**What the response shows:** An unsupported wider claim

**Next teaching action:** Ask which supplied evidence supports the claim and which further observation would be needed.

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Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

## 5. Assess and explain the answers

### Guided and exit questions

Guided: All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures. Qualification: A case passes only all its declared response and trace requirements, not merely a convenient status code. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

### W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision

rule and output.

Teacher explanation: Capstone failure testing checks denied roles, missing evidence, invalid input, timeouts and stale data. Each expected outcome is written before the run. Use a failure matrix linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.

## **W2 Trace the worked case**

Four cases: malformed request, student teacher-guide request, missing register, valid current record. Two deny, one reviews, one supports; no restricted reads occur. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

## **W3 Audit the limit**

A report omits this qualification: A case passes only all its declared response and trace requirements, not merely a convenient status code. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A case passes only all its declared response and trace requirements, not merely a convenient status code. State the observed result separately from the unverified or out-of-scope claim; preserve the original evidence rather than inventing a missing fact.

## **W4 Apply a new case independently**

Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication.

Teacher explanation:  $5/6 \approx 83.333\%$ ; access condition fails and blocks a zero-restricted-read release gate. Do not count final denial as sufficient.

## **W5 Design a revealing follow-up**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked input or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure. Require preserved settings, original outputs and data roles. A case passes only all its declared response and trace requirements, not merely a convenient status code. The new case should test this particular mechanism rather than merely restate the original result.

## **Independent assessment and fresh retry**

W4:  $5/6 \approx 83.333\%$ ; access condition fails and blocks a zero-restricted-read release gate. Do not count final

denial as sufficient. Require intermediate reasoning and the stated scope. Fresh retry: Five cases, all statuses correct, one unauthorised write attempt. Expected reasoning: At most 4/5 contract passes when trace constraints count; action gate fails. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

## Assess independent transfer

Selected case: Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication.

## Selected case answer

Teacher explanation: 5/6 ≈ 83.333%; access condition fails and blocks a zero-restricted-read release gate. Do not count final denial as sufficient.

### Dimension · Evidence required

**Dimension:** Explanation

**Evidence required:** Names the applicable idea or procedure and ties it to the supplied facts.

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**Dimension:** Trace and result

**Evidence required:** Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

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**Dimension:** Limits

**Evidence required:** States the assumptions, missing evidence or conditions under which the result would change.

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Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

## Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

## 6. Connect the project, support and next lesson

For Grounded Learning Service, use this week's independent task as a rehearsal for the final evidence pack: Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication. Keep actual observations separate from expected results, preserve failures, compare a baseline and explain the limits.

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Capstone release review. Preserve the current case record and retrieve its evidence boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

## Plan the delayed return

Suggested return points in this level: ai-6/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

### Record · Teacher evidence

**Record:** Learner code and date

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Concept and case used; prior exposure

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Explanation / trace / limits

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Support and first repair

**Teacher evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Record:** Changed case and delayed outcome

**Teacher evidence:** \_\_\_\_\_

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**Record:** Next teaching action

**Teacher evidence:** \_\_\_\_\_

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Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

## 7. Prepare, teach and assess

Teaching target: Explain the mechanism and terminology of capstone failure testing

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

### Session · Plan · Resources

**Session:** Session 1 · Understand · 45 min

**Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

**Resources:** Sections 1–6; record the brief recall in the workbook.

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**Session:** Session 2 · Practise · 45 min

**Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

**Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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**Session:** Session 3 · Investigate and transfer · 45 min

**Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

**Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

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Before class, solve the supported practice: Four cases: malformed request, student teacher-guide request, missing register, valid current record. Two deny, one reviews, one supports; no restricted reads occur. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

Check every learner before discussion: Use the worked case for Capstone failure testing. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A case passes only all its declared response and trace requirements, not merely a convenient status code.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Make retrieval unavailable as well. Require a prediction and a reason before comparing. Expected result: Return unsupported or unavailable; do not use the factual template. Reason: A template can phrase evidence but cannot replace missing evidence.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

### **Criterion · Not yet · With support · Independent**

**Criterion:** Explanation

**Not yet:** Names an answer without a deciding fact

**With support:** Uses a hint to link fact and decision

**Independent:** Explains the deciding fact in own words

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**Criterion:** Trace or test

**Not yet:** Cannot connect input and result

**With support:** Completes steps with prompts

**Independent:** Traces or tests a changed case accurately

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**Criterion:** Limits and revision

**Not yet:** Claims more than evidence supports

**With support:** Names a limit after a prompt

**Independent:** States a defensible limit and useful next test

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Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

## 8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.